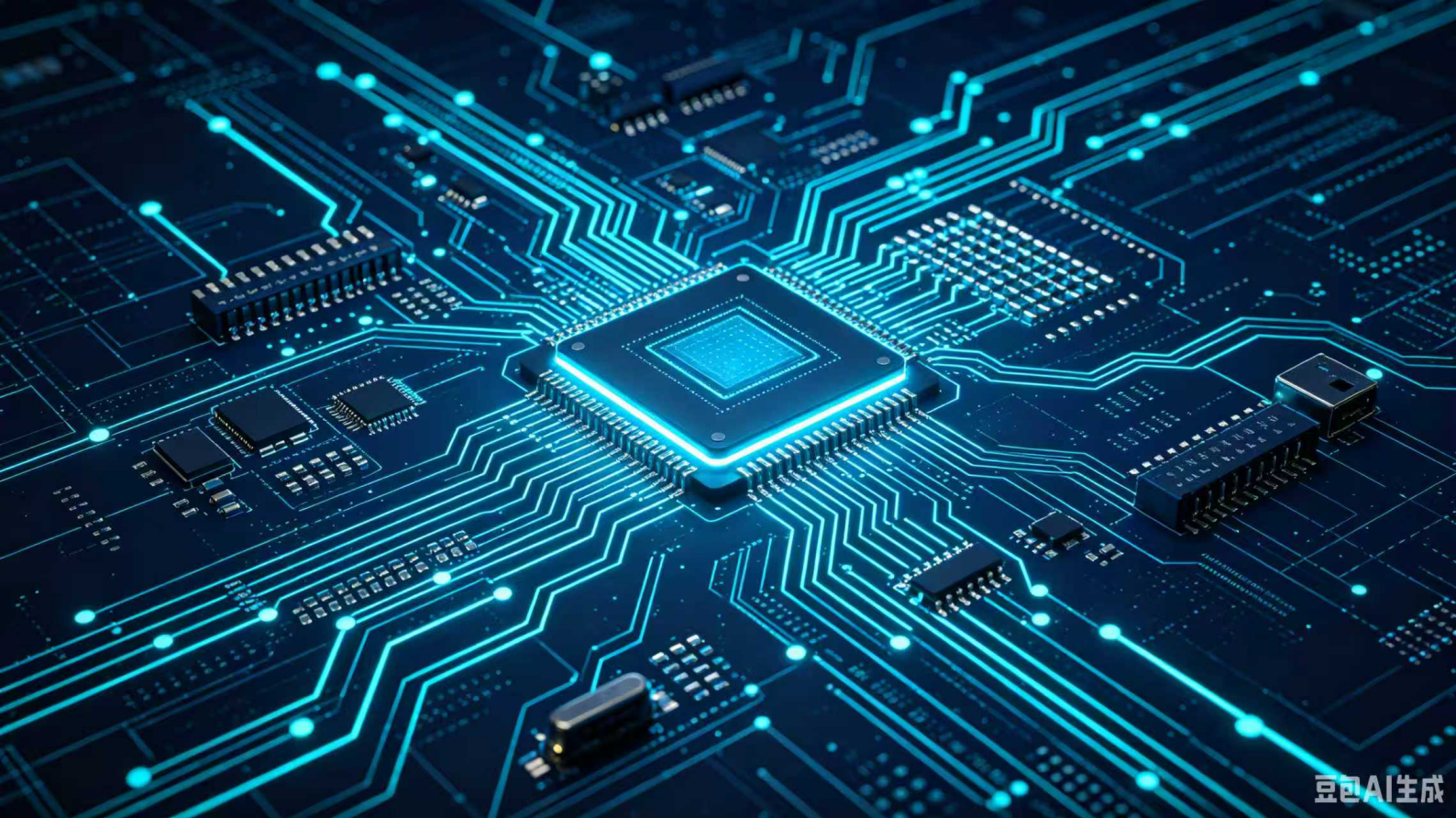




2. Gauss' Law

Symmetry dictates interaction.
– C. N. Yang

Michael Faraday is known to most of us for his contribution to the induction law that is named after him. However, he is also the one who introduced the concept of “field” to physics. Although he did not receive a complete education in mathematics (he barely knew trigonometric functions or integration/differentiation), his remarkable perception of physical ideas made him one of the greatest scientists in history. In his days, Faraday referred to the field as “lines of force”, and he wrote in *Thoughts on Ray Vibrations* (*Experimental Researches in Electricity*, Vol. III, 447-452):

“The view which I am so bold as to put forth considers, therefore, radiation as a high species of vibration in the lines of force which are known to connect particles and also masses of matter together. It endeavors to dismiss the aether, but not the vibration. The kind of vibration which, I believe, can alone account for the wonderful, varied, and beautiful phaenomena of polarization, is not the same as that which occurs on the surface of disturbed water, or the waves of sound in gases or liquids, for the vibrations in these cases are direct, or to and from the centre of action, whereas the former are lateral. It seems to me, that the resultant of two or more lines of force is in an apt condition for that action which may be considered as equivalent to a lateral vibration; whereas a uniform medium, like the aether, does not appear apt, or more apt than air or water.”

By this passage it is apparent that Faraday had not only already developed the concept of field,

understood its importance as a medium of force, argued it was real and different from the ether, but also proposed that radiation (light) is a vibration of the (electromagnetic) field. In 1855, Maxwell restated Faraday's idea mathematically in *On Faraday's Lines of Force*, picturing the lines of force as the flow lines of an incompressible fluid. In that picture the flux through a closed surface counts the sources inside it, which is the content of Gauss' law; for its proof Maxwell refers the reader to Green and to Gauss.

2.1 Electric Flux

The concept of “electric field lines” helps us to understand the abstract concept of electric field. This concept resembles the streamline in the ideal fluid. For incompressible fluid, the flux through a cross section is crucial. Similarly, we have also “flux” of electric field lines. Recall that *the density of the electric field lines in the normal plane is proportional to the magnitude of the electric field at that point*, we see that the number of electric field lines crossing a small patch ΔS , or the **electric flux**, can be defined as:

$$\Delta\Phi = |\mathbf{E}|\Delta S \cos \theta$$

Here θ is the angle formed by the electric field $\mathbf{E}$ and the normal to the patch. We can assign the area a direction, which is just the normal to the patch, such that $\Delta\mathbf{S} = \Delta S\hat{n}$. Thus, we have $\Delta\Phi = \mathbf{E} \cdot \Delta\mathbf{S}$.

In the above, since the patch ΔS is small, we can ignore the variation of $\mathbf{E}$ across ΔS . If the patch is sufficiently large, such that the electric field can no longer be regarded as uniform across it, we can again slice the patch into smaller pieces and sum together. Thus

$$\Phi = \int \mathbf{E} \cdot d\mathbf{S}$$

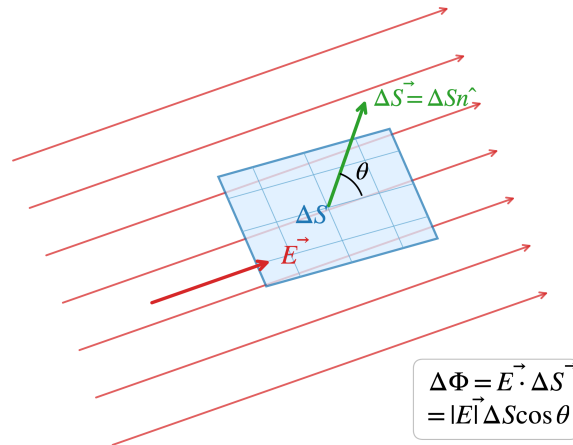


Figure 2.1: Electric flux through a small surface patch ΔS . The flux is $\Delta\Phi = \mathbf{E} \cdot \Delta\mathbf{S} = E\Delta S \cos \theta$, where θ is the angle between $\mathbf{E}$ and the surface normal $\hat{n}$.

Example 2.1: Electric flux through a hemisphere in uniform electric field

We assume the hemisphere has radius R . The angle between the axis of the hemisphere $\hat{z}$ and the electric field $\mathbf{E}$ is α . We take the origin at the center of the hemisphere, and z direction along its axis $\hat{z}$. The electric field lies in the xz plane:

$$\mathbf{E} = E(-\hat{x} \sin \alpha + \hat{z} \cos \alpha)$$

Taking an element of surface at R, θ, ϕ , the surface element is $dS = R^2 \sin \theta d\theta d\phi$, and direction along $\hat{r} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$. Thus the total flux:

$$\begin{aligned} \Phi &= \int_S \mathbf{E} \cdot d\mathbf{S} \\ &= \int R^2 \sin \theta d\theta d\phi \mathbf{E} \cdot \hat{r} \\ &= ER^2 \int_0^{\pi/2} \sin \theta d\theta \int_0^{2\pi} d\phi \times (-\sin \alpha \sin \theta \cos \phi + \cos \alpha \cos \theta) \\ &= ER^2 \int_0^{\pi/2} 2\pi \cos \alpha \cos \theta \sin \theta d\theta \\ &= E\pi R^2 \cos \alpha \end{aligned}$$

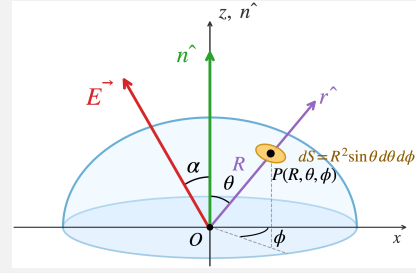


Figure 2.2: A hemisphere of radius R in a uniform electric field $\mathbf{E}$. The surface element $d\mathbf{S}$ at (R, θ, ϕ) points radially outward. Only the component of $\mathbf{E}$ normal to the surface contributes to the flux.

2.2 Gauss' Law

We now look into electric flux through a closed surface, or a **Gaussian surface**. Before that, we have to introduce an important convention: **For a closed surface, a field piercing inward contributes negative flux, whereas an outward piercing field contributes positive flux.** The reason for such a convention will be apparent after we finish the introduction of Gauss' law.

Gauss' law is indeed a very simple relationship between the net electric flux of a **closed** surface (or Gaussian surface) Φ and the net charge enclosed by the surface q_{enc} . It states:

$$\epsilon_0 \Phi = q_{\text{enc}}$$

Or in more detailed form:

$$\epsilon_0 \oint_S \mathbf{E} \cdot d\mathbf{S} = q_{\text{enc}} = \int_V \rho dV \quad (2.1)$$

Here, the integral symbol with circle $\oint_S$ on the left-hand side means the integral takes place on a closed surface S , and the integral symbol $\int_V$ means the integral is on the volume V enclosed by S . Now you can see that the sign convention defined previously simply means electric field lines are sent out of positive charges and are received by negative charges. You can imagine the positive charges as out-going source, and the negative charges as in-coming drain. To balance the flux and keep the distribution constant in time, the infinity can also be source or drain. The electric field is the flux sent out from the source to the drain, and cannot be absorbed or created anywhere else. Thus, the amount of net flux through a closed surface is definitely proportional to the net source (meaning the total amount of source charge minus drain charge) inside the surface. The source and drain outside the closed surface change the distribution of the flux, but do not contribute to the net

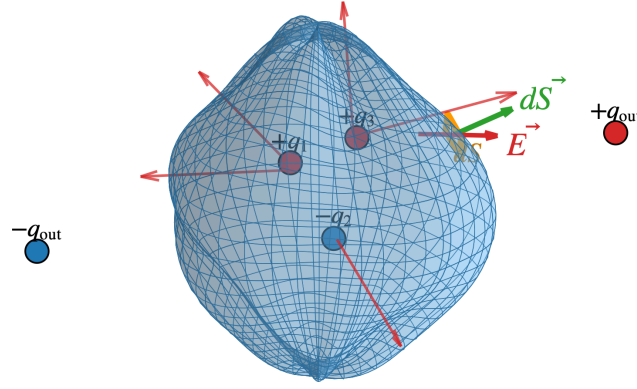


Figure 2.3: A closed Gaussian surface S enclosing charges $+q_1, -q_2, +q_3$. Charges outside the surface ($\pm q_{\text{out}}$) contribute to $\mathbf{E}$ locally but their net flux through S is zero. The surface element $d\mathbf{S}$ and the electric field $\mathbf{E}$ at a point on the surface are shown.

flux through the surface. In terms of the electric field, this means **the electric field due to charges outside the Gaussian surface has zero contribution to the net flux through the surface.**

Example 2.2: Gauss' law applied to point-charge systems.

For the simplest case, we verify Gauss' law for a single point charge $q > 0$ in an empty space. Assume that q is located at the origin, and that we use a spherical surface of radius R around it as the Gaussian surface. According to Coulomb's law, the electric field is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

where r and $\hat{\mathbf{r}}$ are the norm and direction of the position vector $\mathbf{r}$, respectively.

Since the center of the Gaussian surface is also located at the point-charge position, $\hat{\mathbf{r}}$ is always perpendicular to the Gaussian surface as well. For a spherical surface of radius R , the area element $d\mathbf{S} = \hat{\mathbf{R}} R^2 d\Omega$, where $d\Omega$ is the solid angle spanned by the area element and $\hat{\mathbf{R}}$ is the radial direction vector. Thus, we simply have $\hat{\mathbf{r}} \cdot d\mathbf{S} = R^2 d\Omega$:

$$\Phi = \oint_S \mathbf{E} \cdot d\mathbf{S} = \oint_S \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} R^2 d\Omega = \frac{q}{4\pi\epsilon_0} \oint_S d\Omega = \frac{q}{4\pi\epsilon_0} \cdot 4\pi = \frac{q}{\epsilon_0}$$

This verifies Gauss' law for this simplest case.

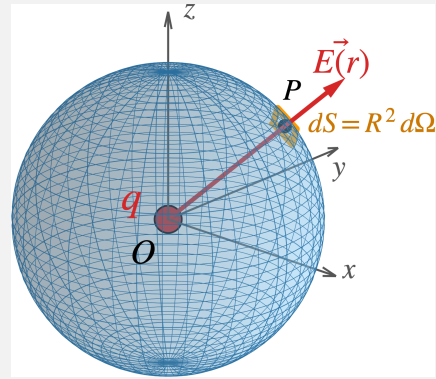


Figure 2.4: A point charge q at the center of a spherical Gaussian surface of radius R . At point P on the sphere, the surface element $d\mathbf{S} = R^2 d\Omega \hat{\mathbf{R}}$ and the electric field $\mathbf{E}$ are both radial, so $\mathbf{E} \cdot d\mathbf{S} = E R^2 d\Omega$.

There are two remarks we want to make:

1. Firstly, Gauss' law appears to be derived from electrostatic field here, but it is in fact more fundamental than that. In fact, it is one of Maxwell's four equations, so it also applies to dynamic fields or charges in motion.
2. Secondly, the specific form here is sometimes referred to as "Gauss' law in vacuum". However, this may be confusing. There is nothing to be changed for Gauss' law in medium, if

$\mathbf{E}$ refers to the electric field at the microscopic scale and all microscopic charge is included. However, the microscopic field varies abruptly in a medium and is not very useful, thus we need to consider the averaged field $\langle \mathbf{E} \rangle$. Then Gauss' law for this averaged field, when written in terms of the free charge alone, takes a different form, which is sometimes referred to as "Gauss' law in medium". We shall detail this later on when we try to derive the expression for electrostatic field in a medium.

2.3 Application of Gauss' Law in Highly Symmetric Cases

One of the many applications of Gauss' law is to simplify the calculation of the electric field in highly symmetric ideal cases. It is, however, important to point out that Gauss' law also holds in non-ideal nonsymmetric cases, as argued above.

2.3.1 Spherical Symmetry

By spherical symmetry, we mean that the spatial distribution of the electric charge depends on its distance from the origin r only, without any angular dependence. In such a system, it is simplest to take spherical coordinates (r, θ, ϕ) .

Example 2.3: uniformly charged spherical shell again.

We calculate the electric field generated by a uniformly charged spherical shell of radius R . The charge carried by the shell is $Q > 0$ and the thickness of the shell can be ignored. There are two situations:

1. Outside the sphere.

In this case, we take a Gaussian surface of radius $r > R$, whose center coincides with the center of the shell. Applying Gauss' law, we have:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}$$

Again, due to symmetry, the magnitude of $\mathbf{E}$ must be the same on this surface, and its direction must be radial. Therefore

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

and its direction is along $\hat{r}$.

2. Inside the sphere.

Take a concentric Gaussian sphere of radius $r < R$. It encloses no charge, so the flux through it is zero. By symmetry $\mathbf{E}$ is radial and has the same magnitude at every point of this sphere, so the flux equals $4\pi r^2 E$. Therefore $E = 0$ everywhere inside the shell.

Thus, for a spherical charge distribution, the electric field at r is that of a point charge at the center equal to the total charge inside the sphere of radius r .

2.3.2 Planar Symmetry

By planar symmetry, we mean that the electric charge density $\rho(z)$ depends on z only, where (x, y, z) is the Cartesian coordinate.

Example 2.4: uniformly charged infinitely large plate.

The surface charge density of the plate is $\sigma > 0$, and the thickness of the plate can be ignored.

In this case, we take a cylindrical Gaussian surface, whose two ends are parallel to and symmetrically placed with respect to the plate. Symmetry considerations dictate the electric field $\mathbf{E}$ is perpendicular to the plate and the magnitude of $\mathbf{E}$ depends on (at most) $|z|$. Reflecting the plate in its own plane leaves it unchanged, so $E_z(-z) = -E_z(z)$: the field points away from the plate on both sides or toward it on both sides, and the two end surfaces contribute equally. Therefore, the side of the cylinder does not contribute to Φ , and each end surface contributes $\Phi_{\text{cap}} = \int \mathbf{E} \cdot d\mathbf{S} = ES$, because $\mathbf{E}$ is perpendicular to the plate, and likewise perpendicular to the end surfaces. The enclosed total charge is $q = \sigma S$, therefore, Gauss' law becomes:

$$2ES = \frac{\sigma S}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

whose direction is perpendicular to the plate, pointing away from the plate for $\sigma > 0$ and toward it for $\sigma < 0$:

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \text{sgn}(z) \hat{z}$$

where $\text{sgn}(z)$ takes the sign of z : $\text{sgn}(z) = z/|z|$.

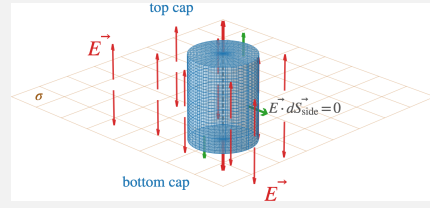


Figure 2.5: A cylindrical Gaussian surface (pillbox) straddling an infinite charged plate with surface charge density σ . The end caps of area S are parallel to the plate. By symmetry, $\mathbf{E}$ is uniform over each cap and perpendicular to it, so the flux through each cap is ES .

2.3.3 Cylindrical Symmetry

By cylindrical symmetry, we mean that the charge density $\rho(r, \phi, z)$ depends on r only, where (r, ϕ, z) are cylindrical coordinates and r is the distance from the axis.

We leave this case to the exercises (Exercise 3).

2.4 Exercises

- An electric field is given by $\mathbf{E} = y^2\hat{i} + (y - x^2)\hat{j} + \frac{xy}{z}\hat{k}$. At a point $\mathbf{r} = (3, 2, 1)$, please calculate the *local* flux per area if the local area vector is:
 - $\hat{i}dS$
 - $\hat{j}dS$
 - $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})dS$
- A spherically symmetric charge density $\rho(r)$ produces an electric field that points radially outward and has the same magnitude E at every $r > 0$. Please give the expression of $\rho(r)$.
- Consider an infinitely long cylinder of radius R and a uniform charge density $\rho_0 > 0$. Use Gauss' law to determine the magnitude of the electric field at a distance r from the central axis of the cylinder. You should give the expressions of E for both $r \leq R$ and $r > R$.
- Two uniformly charged spherical **shells** are both centered at the origin, whose radii are R_1 and R_2 ($R_1 < R_2$). Given that the surface charge density is $+\sigma$ ($\sigma > 0$) for the outer shell, and that the electric field is zero for any point located outside the outer shell, please calculate the following quantities:
 - The surface charge density of the inner shell
 - The magnitude of the electric field for a point located in between the two shells ($R_1 < r < R_2$)
 - The magnitude of the electric field inside the inner shell ($r < R_1$)
- Two uniformly charged **balls**, of radii R_1 and R_2 and charge densities $-\rho$ and $+\rho$ with $\rho > 0$, overlap: their centers are a distance $d < R_1 + R_2$ apart, and in the overlap region the two charge densities add (see figure below). Let $\mathbf{d}$, with $|\mathbf{d}| = d$, be the vector from the center of the $-\rho$ ball to the center of the $+\rho$ ball. Please calculate both the magnitude and the direction of the electric field in the overlap region, in terms of ρ and $\mathbf{d}$.

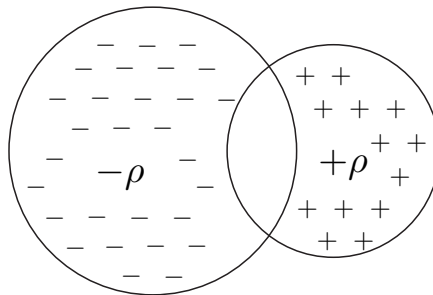


Figure 2.6: Two uniformly charged balls.

- A point charge $q > 0$ and an imaginary cube of side a are given. The normal of every face of the cube points out of the cube.
 - The charge sits at the center of the cube. Find the electric flux through each face.

- (b) The charge now sits at one corner of the cube. Find the electric flux through each of the six faces. (Hint: consider the eight identical cubes that share this corner.)
7. A point charge $q > 0$ lies on the axis of a flat disk of radius R , at a distance $d > 0$ from the center of the disk. The normal of the disk points away from q .
- (a) The sphere of radius $\sqrt{d^2 + R^2}$ centered on q passes through the rim of the disk. The disk and the part of this sphere on the far side of the disk from q together enclose a region that contains no charge. Use Gauss' law for this closed surface to show that the flux through the disk equals the flux through that part of the sphere, and hence find the flux through the disk. Check your answer when $R \gg d$ and when $R \ll d$.
- (b) The charge is now at the center of a closed cylinder of radius R and length $2d$ whose axis passes through q . Without evaluating any new integral, find the flux through the curved side of the cylinder.
8. Two point charges, each $q > 0$, are fixed at $(0, 0, a)$ and $(0, 0, -a)$, where $a > 0$. Let S be the sphere of radius $R > a$ centered at the origin, with its normal pointing outward.
- (a) Find the net electric flux through S .
- (b) Calculate the electric field at the two points $(R, 0, 0)$ and $(0, 0, R)$ of S .
- (c) From (a), find the average outward component of the field over S ,

$$\langle E_n \rangle = \frac{1}{4\pi R^2} \oint_S \mathbf{E} \cdot d\mathbf{S},$$

and compare it with the outward components found in (b). Can Gauss' law alone give the electric field at every point of S ? Why could it give the field of a uniformly charged spherical shell?

9. In a simple model of a p-n junction, the charge density depends only on z : $\rho(z) = -\rho_0$ for $-a < z < 0$, $\rho(z) = +\rho_0$ for $0 < z < a$, and $\rho(z) = 0$ for $|z| > a$, where $\rho_0 > 0$. The charged layers extend infinitely in the x and y directions, and there is no other charge.
- (a) Treat the charge as a stack of thin charged plates, and use the field of an infinitely large charged plate to show that $\mathbf{E} = 0$ for $|z| > a$. Explain why a pillbox placed symmetrically about $z = 0$, as used for a single plate, gives no information here.
- (b) Using a pillbox with one end in the region $z < -a$, find $\mathbf{E}(z)$ for $-a < z < a$. Where is the magnitude of the field largest, and what is its value there?
10. Suppose Coulomb's law were slightly different, so that a point charge $q > 0$ at the origin produced the field

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{r_0}{r}\right)^\delta \hat{r},$$

where r_0 is a fixed length and δ is a small positive number.

- (a) Calculate the flux of this field through a sphere of radius R centered on q . Is the result consistent with Gauss' law?
- (b) Two concentric spheres of radii $R_1 < R_2$, centered on q , together form a closed surface that encloses the region $R_1 < r < R_2$ and no charge. Its normal points out of that region, that is, toward q on the inner sphere. Calculate the net flux through this closed surface. Which property of the flux, used throughout this chapter, fails? What does this tell you about the exponent in Coulomb's law?